

Norway

SLEIPNER

Volve F

F-9

F-9

Design: F-9

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-9	North Reference:	Grid
Wellbore:	F-9	Survey Calculation Method:	Minimum Curvature
Design:	F-9	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site		Volve F, 15/9				
Site Position:		Northing:	6,478,563.53 m	Latitude:	58° 26' 29.8068 N	
From:	Map	Easting:	435,050.02 m	Longitude:	1° 53' 14.9285 E	
Position Uncertainty:		0.00 m	Slot Radius:	13.200 in	Grid Convergence:	-0.95 °

Well	F-9					
Well Position	+N/-S	-1.96 m	Northing:	6,478,561.57 m	Latitude:	58° 26' 29.7442 N
	+E/-W	1.11 m	Easting:	435,051.13 m	Longitude:	1° 53' 14.9992 E
Position Uncertainty		0.00 m	Wellhead Depth:	91.00 m	Water Depth:	91.00 m

Wellbore	F-9				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	MAGNETICREFERENCE	25.11.2007	-2.25	71.61	50,362

Design	F-9				
Audit Notes:					
Version:	1.0	Phase:	ACTUAL	Tie On Depth:	145.90
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)	
	145.90	-1.96	1.11	77.00	

Survey Program	Date	31.05.2010			
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description	
218.50	902.60	F9 17 1/2in Surveys (F-9)	Keeper, stat	Gyro Tool from SDC	
968.02	1,048.86	F-9 12 1/4in MWD (F-9)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)	

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
145.90	0.00	0.00	145.90	-1.96	1.11	0.00	0.000	0.000	0.000	
218.50	0.26	197.21	218.50	-2.11	1.07	-0.08	0.107	0.107	0.000	
259.00	0.18	255.31	259.00	-2.22	0.98	-0.19	0.167	-0.059	43.037	
299.50	0.35	240.89	299.50	-2.29	0.81	-0.38	0.134	0.126	-10.681	
340.00	0.44	34.96	340.00	-2.23	0.79	-0.38	0.570	0.067	114.126	
394.30	0.35	344.16	394.30	-1.90	0.86	-0.23	0.193	-0.050	-28.066	
420.40	0.35	352.69	420.40	-1.74	0.83	-0.23	0.060	0.000	9.805	
460.60	0.35	35.05	460.60	-1.52	0.89	-0.12	0.189	0.000	31.612	
500.80	0.35	36.99	500.80	-1.32	1.03	0.06	0.009	0.000	1.448	
540.90	2.90	101.32	540.88	-1.42	2.10	1.08	2.070	1.908	48.127	

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Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
580.40	3.87	101.23	580.31	-1.88	4.39	3.21	0.737	0.737	-0.068
620.80	5.63	104.05	620.57	-2.62	7.65	6.21	1.318	1.307	2.094
661.50	7.13	89.02	661.02	-3.06	12.11	10.46	1.646	1.106	-11.079
701.80	8.71	83.48	700.93	-2.67	17.64	15.94	1.305	1.176	-4.124
742.20	11.70	82.34	740.69	-1.78	24.74	23.06	2.225	2.220	-0.847
782.60	11.17	79.17	780.29	-0.50	32.64	31.05	0.610	-0.394	-2.354
823.00	11.44	73.46	819.90	1.38	40.33	38.96	0.855	0.200	-4.240
863.30	11.61	68.27	859.39	4.02	47.93	46.96	0.782	0.127	-3.864
902.60	10.03	74.43	897.99	6.40	54.90	54.29	1.493	-1.206	4.702
968.02	9.52	74.93	962.46	9.33	65.61	65.38	0.237	-0.234	0.229
1,008.50	9.41	75.46	1,002.39	11.03	72.05	72.04	0.104	-0.082	0.393
1,048.86	9.61	76.13	1,042.20	12.67	78.51	78.70	0.170	0.149	0.498
1,083.00	9.61	76.13	1,075.86	14.04	84.04	84.40	0.000	0.000	0.000

Design Targets									
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (m)	+N/-S (m)	+E/-W (m)	Northing (m)	Easting (m)	Latitude	Longitude
- hit/miss target									
- Shape									
Rev Utsira Top v2 07110	0.00	0.00	874.90	6.54	100.01	6,478,570.06	435,150.00	58° 26' 30.0716 N	1° 53' 21.0856 E
- actual wellpath misses target center by 48.42m at 887.93m MD (883.56 TVD, 5.64 N, 52.38 E)									
- Point									
Utsira Top F9	0.00	0.00	879.90	6.48	50.00	6,478,570.00	435,100.00	58° 26' 30.0429 N	1° 53' 18.0031 E
- actual wellpath misses target center by 2.02m at 883.96m MD (879.66 TVD, 5.40 N, 51.69 E)									
- Point									
Rev Utsira Top 071106	0.00	0.00	879.90	6.54	50.00	6,478,570.06	435,100.00	58° 26' 30.0449 N	1° 53' 18.0031 E
- actual wellpath misses target center by 2.05m at 883.96m MD (879.66 TVD, 5.40 N, 51.69 E)									
- Point									
Utsira Base F9	0.00	0.00	1,058.90	1.52	409.62	6,478,565.04	435,459.50	58° 26' 30.0742 N	1° 53' 40.1716 E
- actual wellpath misses target center by 326.26m at 1083.00m MD (1075.86 TVD, 14.04 N, 84.04 E)									
- Point									

Casing Points					
Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)	
220.80	220.80	30"	30.000	36.000	
909.80	905.08	13 3/8"	13.375	17.500	
1,064.00	1,057.13	9 5/8"	9.625	12.250	

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Design Annotations					
Measured Depth (m)	Vertical Depth (m)	Local Coordinates			
		+N/-S (m)	+E/-W (m)	Comment	
1,083.00	1,075.86	14.04	84.04	Project to TD	

Checked By: _____	Approved By: _____	Date: _____
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